

OH_01: Order Basic Payment Bundle - Payment Account - Order Fulfilment

Public Digital PSO Solution Architect

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1 Introduction

1.1 Purpose and Scope

This document extends Epic **OC_01 - Order Basic Payment Bundle - Order Capture & Product Inventory**, by enabling customers to apply for a new **Basic Payment Bundle with a Payment Account** only and activates the Payment Account and the Bundle Package in ABN AMRO external systems.

Activating a Debit card and fulfil it, will be solutioned in Epic OH_02.

Line Of Business

Payment Account

Customer Segment

B2C (Business-to-Consumer) Banking Customer

Channels

Internet Banking (Self-Service)

1.2 Business Needs

As an existing customer, once I applied via Internet Banking site for a new Basic Payment Account Bundle and signed on the required documents, I would like to have my **Payment Account** and **Bundle Agreement** activated.

1.3 Application Involved

Application	Description	Provider	Impacted
Catalog ONE	Amdocs Product Catalog (Catalog ONE) enables the management of offers, products, and accessories. It defines product characteristics, rules, pricing policies, and eligibility conditions. The Basic Payment Bundle will be created in Catalog One to support the Bundle ordering for ABN AMRO.	Amdocs	Yes

Application	Description	Provider	Impacted
Digital ONE	Amdocs Digital ONE provides a set of value-based, self-contained business functions which support diverse commerce and care workflows. The following business functions will be used in ABN AMRO banking solution: In Order Handling - New Business process map will be introduced to activate IBAN and agreement in external systems.	Amdocs	Yes
ABN AMRO Systems	CRC - Centrale Reeking Courant ABN AMRO banking system which enables activation and reservation of the iBAN. APA - Agreement Proposition Administration ABN AMRO banking system which handles the packages of the customer.	ABN AMRO	No

1.4 Related Document

Document ID	Type	Document Name
OC_01		Order Basic Payment Bundle - Order Capture & Product Inventory
OC_02		Order Basic Payment Bundle - iBAN Reservation
APA		APA YAML ¹
CRC		CRC YAML ²
APA Integration from D1		APA - /POST (Product Bundle Agreement API) ³

¹ https://amdocs.sharepoint.com/:u:/r/sites/ABN-AMRO2/Shared%20Documents/AMRO%20Project/Solution/External%20Systems/APA%20-%20ProductBundleAgreementAPI/ProductBundleAgreementAPI_v2.yaml?csf=1&web=1&e=dkdF1e

² <https://confluence/download/attachments/516556717/Current%20Account%20API.yaml?api=v2&modificationDate=1740659064732&version=1>

³ <https://confluence/pages/viewpage.action?pageId=529564694>

Document ID	Type	Document Name
CRC Integration from D1		POST /createCurrentAccount (Active IBAN) ⁴

1.5 Assumptions

ID	Item	
01	Product Inventory	Although Digital ONE stores the product inventory and activates it in D1, the master record of customer-owned products remains in ABN AMRO system. Any queries on PI should be performed against the master system of record.
02	Charging	ABN AMRO systems will be responsible for notifying the ABN AMRO charging system.
03	CRC Integration	The integration between D1 and CRC is sync. Once CRC return success status, it is CRC responsibility to handle failures/ functionalities post this returned status. (functionalities such as notifying P&C or other systems). The success result from CRC will derive the successful completion of the step in D1 and the continuation to the next step in the map.
04	APA Integration	The integration between D1 and APA is sync. Once APA return success status, it is APA responsibility to handle failures/ functionalities post this returned status. (functionalities such as notifying P&C or other systems). The success result from APA will derive the successful completion of the step in D1 and the continuation to the next step in the map.

⁴ <https://confluence/pages/viewpage.action?pageId=516556717>

ID	Item	
05	One Customer	• Customer Requestor (Creator): The customer who logs in and places the order, either for himself or on behalf of another customer. • Receiver (BC ID): The customer to whom the product will belong. The receiver can be either the customer requestor or a different customer. A Product Order and Product Inventory are owned by a single customer receiver. For this EPIC, the receiver is assumed to be the customer requestor. A bundle within a Product Order is associated with one customer (receiver)
06	Handled in future epics	Debit card activation will be handled in Epic OH_02 Notification to customer when the order is completed will be handled in future epic.
07	Single PA and single package in the order	Single Payment Account/Package is activated as part of the order. In case additional PA are required as part of the order, it will be handled in future epic.
08	PA Product ID	A single product ID will be defined in AMRO for Payment Account, either when its within or outside of a bundle.

1.6 Terms & Definitions

Term	
C1 / CatalogONE	Amdocs CatalogONE is an online product catalog based on TM Forum standards. It enables the management of commercial offerings and product definitions.
D1 / DigitalONE	Amdocs DigitalONE consists of a set of functional capabilities used to handle the ordering flow, including order capture and fulfillment.
OC	Amdocs D1 Order Capture (OC) solution consists of a set of functional capabilities used to handle the negotiation of the order before submission
AOM	The Amdocs Order Management (AOM) solution consists of a set of functional capabilities used to handle the order after submission and for Amdocs Order Fulfillment. Also referred as Order Handling.

2 High Level Solution

2.1 Overview

This document describes the fulfillment of the order including **activating the Payment Account** (iBAN) that was reserved, **and** the **New Payment Bundle** in the relevant external systems.

2.1.1 Amdocs Order Management (AOM)

The order handling process will be responsible for fulfillment of the Basic Payment Account Bundle order.

The following new activities, will be incorporated into the order fulfillment flow:

- **Activate Payment Account**– Activate the **Payment Account (iBAN)** by calling the external API of ABN AMRO **CRC** for triggering the fulfillment of the PA.
- **Activate Package of the bundle**– Activate the **bundle package** calling the external API call of ABN AMRO **APA** for triggering the activation of the Bundle Package. Store the returned instance ID on the product in D1.

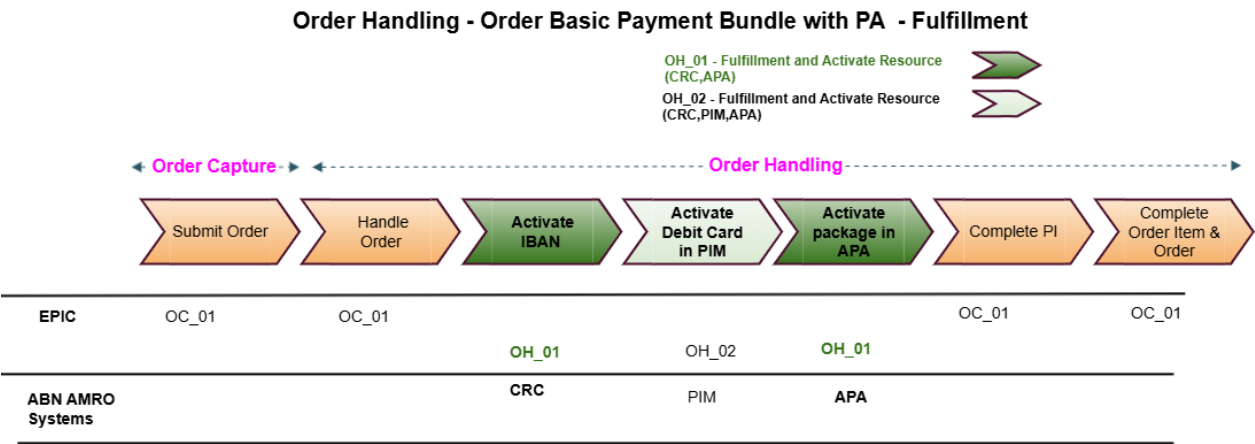


Eligibility check during fulfillment

As the order is validated during the submit order stage and the signoff activity will be handled in future epic, no eligibility check will be performed as part of fulfillment.

2.2 Business Flows

2.2.1 Flow Diagram - High Level



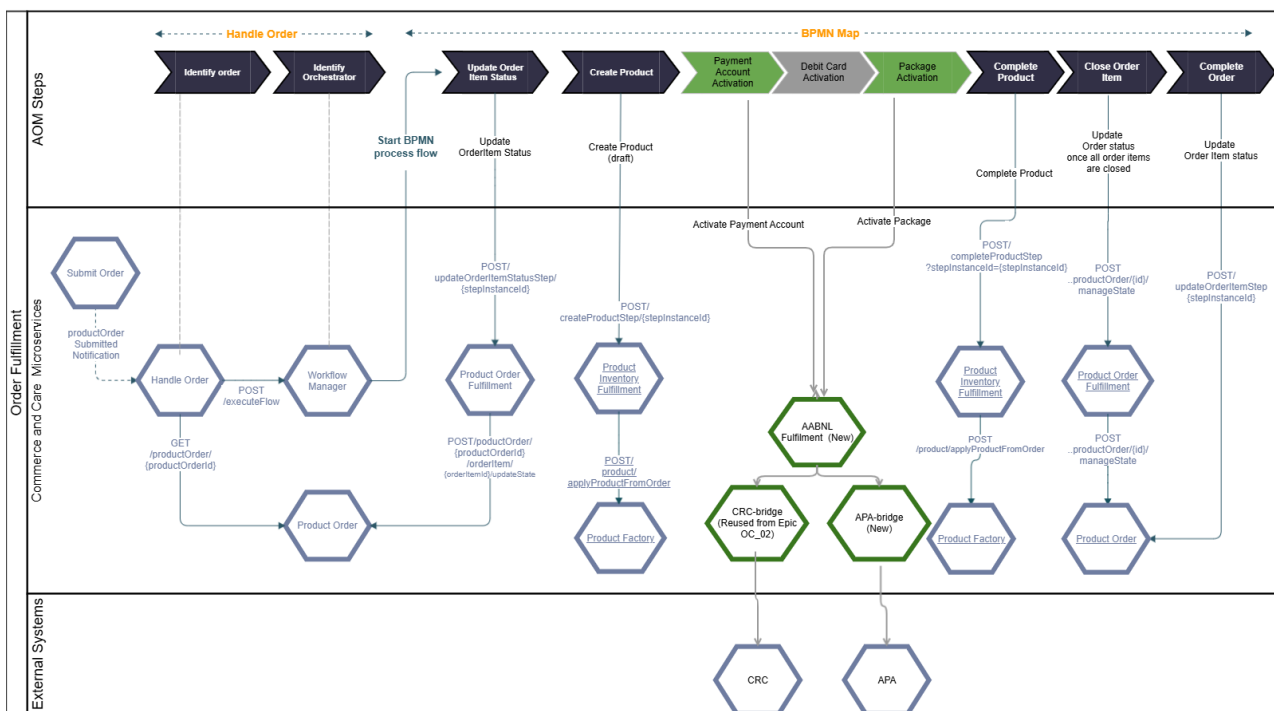
2.3 Business Use Cases

Use Case ID		
#01	The customer orders a Basic Payment Account Bundle, with Payment Account only and submits the order for fulfillment	The Order is fulfilled successfully . The Payment Account (iBAN) and the package are activated successfully .

3 D1 Order Handling- Capabilities

3.1 Flow Diagram

3.1.1 Diagram



3.1.2 Action Integration Point

Step	Commerce Sub Domain	Description
Payment Account Activation	AABNL Fulfillment	Activate the IBAN through CRC Integration

Step	Commerce Sub Domain	Description
Bundle Package Activation	AABNL Fulfillment	Activate the Package through APA Integration

3.2 Solution Details

3.2.1 Flow Description

New Business Processes (BPM) will be created for the new Bundle and another one for the product of the Payment Account - per the order items of the order.

The following new steps will be added to the Business Process Map for supporting order of Basic Payment Account Bundle:

Step	Relevant for Order Item Fulfillment Map	Mandatory/ Conditional	Description
Activate Payment Account	Add Payment Account	Mandatory	The Payment Account Management system (CRC) activates the iBAN in the order
Activate Package	Add Bundle Add Payment Account	Mandatory	The Package Account Management system (APA) activates the Bundle Package in the order. The APA step will be common and will be triggered only once for the bundle and will include both the Payment Account and the package.
Update Order Item	Add Bundle	Mandatory	Update the order item of the bundle with the returned instance ID of APA

3.2.2 Activate Payment Account (iBAN)

3.2.2.1 Overview

Operation	/POST (Product Bundle Agreement API) ⁵
D1 Microservice	AABNL Fulfillment MS
Overview Details	Activate a payment Account (iBAN)

3.2.2.2 Description

A new **DigitalONE** operation will be introduced to activate a payment Account within the product order item.

Supported Resource Type: IBAN with CRC Integration

As part of Epic OC_02 the iBAN resource was defined in C1.

	Request to CRC API	Description	D1 Source of data
1	customerId	The receiver	The customer ID of the order
2	currencyCode	EURO (get from the catalog the default currency)	From the definition in C1
3	productid	Product ID of the PA	From the definition in C1 The same Product ID of the PA will be used when the PA is within or outside a bundle - 437.
4	iban	iban	From the resource ID of the order item

CRC Activation Process

⁵ <https://confluence/pages/viewpage.action?pageId=529564694>

- Based on the catalog definition of the Payment Account offer, get the characteristics of the "Fulfillment Administration" to determine to which system to call (and to which bridge MS to call).
- For IBAN Activation, **DigitalONE** will integrate with the **CRC activation API**
- The iBAN will be the resource ID retrieved from the order item. where the resource type = iBAN

CRC will return the following codes

Returned code	Description
201	Successfully created current account
400	Functional error And the Errors Array that will include a Detailed Error list such as : "CUSTOMER_ID_INVALID", "IBAN_ALREADY_EXISTS"..
500	Internal error occurred, the Errors Array will include the error INTERNAL_SERVER_ERROR "Internal server error occurred"

3.2.2.3 Integration Point

Functionality	Application	Operation	Description	Provider	Consumer	Interface Type	Mode	Integration Direction
Activate	CRC	POST / createCurrentAccount ⁶	Activate resource	ABN AMRO	Amdocs	REST	Syn c	Southbound

⁶ <https://confluence/pages/viewpage.action?pageId=516556717>

3.2.3 Activate Package

3.2.3.1 Overview

Operation	/POST (Product Bundle Agreement API) ⁷
D1 Microservice	AABNL Fulfillment MS
Overview Details	Activate a package in APA

3.2.3.2 Description

A new **DigitalONE** operation will be introduced to activate a package in APA.

APA Activation Process

- Based on the catalog definition of the Bundle offer, get the characteristics of the "Fulfillment Administration" to determine to which system to call (and to which bridge MS to call).
- For Package Activation, **DigitalONE** will integrate with the **APA activation API**
- Once successfully activated, the **External APA Package ID** will be persisted in the **Payment Account** product.

Operation Logic:

Populate the input according to the following:

	Request to APA API	Description	D1 Source of data
1	productBundledId	Product ID of the bundle offer	From the definition of the bundle in C1 (this is the offer ID of the package in AMRO)
2	agreementPartyClusterId	Customer - Receiver BC ID	The customer ID of the order
3	startDate	Submission date of the order	The signed date (in this epic will use the order submission date)
4	linkedProductAgreement	Array with single record	

⁷ <https://confluence/pages/viewpage.action?pageId=529564694>

	Request to APA API	Description	D1 Source of data
5	linkedProductAgreement.productAgreementId	Payment Account product ID - instance of the Payment Account Last 10 digits of the iBAN will be passed	From the resource ID of the order item - last 10 digits
6	linkedProductAgreement.startDate	Submission date	The signed date (in this epic will use the order submission date)
7	linkedProductAgreement.productId	Product ID of the PA offer	From the definition of the PA in C1 (this is the offer ID of the PA in the external catalog)
8	linkedProductAgreement.productSegmentId	The reference number of the segment eg 20 for debit card	From the definition of the PA in C1 (this is the segment ID of the PA in the AMRO)

	Response from APA API	Description	D1 Source of data
1	productBundleAgreementId	instance number of the bundle offer of APA	Should be stored as an external ID on the product of the bundle in D1.

APA will return the following codes

Returned code	Description
201	Agreement Created Successfully
400	"Client Side / Functional Error" Detailed Error List such as: PRODUCT_BUNDLE_ID_MISSING, PRODUCT_BUNDLE_ID_INVALID, LINKED_PRODUCT_AGREEMENTS_MISSING..

Returned code	Description
401	"Authentication Error" Detailed Error List: USER_UNAUTHENTICATED
403	"Authorization Error" Detailed Error List: USER_NOT_AUTHORIZED
500	Internal error occurred, the Errors Array will include the error: INTERNAL_SERVER_ERROR "Internal server error occurred"

3.2.3.3 Integration Point

Functionality	Application	Operation	Description	Provider	Consumer	Interface Type	Mode	Integration Direction
Activate	APA	/POST (Product Bundle Agreement API) ⁸	Activate Package	ABN AMRO	Amdocs	REST	Sync	Southbound

⁸ <https://confluence/pages/viewpage.action?pageId=529564694>

3.3 Acceptance Criteria

3.3.1 Sunny Day Scenario

Test Name	GIVEN	WHEN	THEN
As part of fulfillment, Activating IBAN that was previously reserved	Given the iBAN that is passed in the request to CRC API was reserved	When CRC processes the activation request successfully (returns code 201)	Then it should return successfully iBAN activation result AND the step is completed successfully and the next step starts.
As part of fulfillment, Activating the package with the Payment Account that was successfully activated	Given the input required parameters are passed in the request to APA API	When APA processes the activation request successfully (returns code 201)	Then it should return successfully package activation result Package ID value received from APA should be persisted in the Bundle product

3.3.2 Rainy Day Scenario

Test Name	GIVEN	WHEN	THEN
PA Activation step Fails Due to internal server error	Given the customer has requested a new payment account and the Order Handling of PA Activation step fails	When fulfillment step fails due to CRC API returning internal server error 500	Then system should Retry the call according to the configuration. After the retry attempts exhausted, the system should change the step to fallout.
PA Activation step Fails Due to Functional error	Given the customer has requested a new payment account and the Order Handling of PA Activation step fails	When fulfillment step fails due to CRC API returning Functional error 400	Then system should change the step status and the order item status and the Order status.

Test Name	GIVEN	WHEN	THEN
PA Activation succeeds but the Package activation step fails	Given the customer has requested a new payment account and the Order Handling of PA Activation step succeeds but the Package Activation step fails	When the PA has been activated successfully for this account And the system attempts to activate the package via APA Integration And the APA Integration API fails to process the package activation due to returned Internal error code 500 or Authentication Error 401 or Authorization Error 403	Then The product will not be completed until APA step is fulfilled successfully. The system should Retry the call according to the configuration. After the retry attempts exhausted, the system system should change the step status and the order item status and the Order status.
PA Activation succeeds but the Package activation step fails	Given the customer has requested a new payment account and the Order Handling of PA Activation step succeeds but the Package Activation step fails	When the PA has been activated successfully for this account And the system attempts to activate the package via APA Integration And the APA Integration API fails to process the package activation due to returned Client Side / Functional Error 400	Then The product will not be completed until APA step is fulfilled successfully. System system should change the step status and the order item status and the Order status.

4 ABN AMRO system Integration

Following are the external systems used by Amdocs Ordering Systems:

Application		Provider OAR ID	Consumer	Integrations Direction	Description
CRC Centrale Rekening Courant	Reserve and Activate the IBAN Number	AAB.SYS.1175	Amdocs OC & AOM	South bound	Central administration for current accounts. In OC, Reserve IBAN for the payment bank account (random number) In AOM, activate the IBAN payment account
APA Agreement Proposition Administration	Package Fulfillment	AAB.SYS.017617	Amdocs AOM	South bound	Only the package product is fulfilled. fulfilment includes only the relation with the packaged products, no other details about the packaged products.

5 Operational Aspects

5.1 Non-Functional

5.2 Authorization & Security

5.3 Error Handling

Errors will be logged in the log.

Any additional Error Handling should be solutioned in future epic.

5.4 Performance

TBD

5.5 Migration

Not Applicable.

Legacy Orders which was not initiated by Amdocs Ordering will not be migrated to Amdocs Ordering System.

6 Appendix

Not Applicable.

6.1 Release Notes

Version	Author	Date	Comments
0.1	Dana Roffe	5-Dec-2024	First Release
0.2	Mirit Danieli	10-Mar-2025	

7 Feature OH_01_Fullfilment_PA and Package

7.0.1 Related Document

Document ID	Type	Document Name
Epic		OH_01: Order Basic Payment Bundle - Payment Account - Order Fulfilment (see page 4)
Camunda Step Integration Patterns		Camunda Step Integration Pattern⁹
APA		APA YAML¹⁰
CRC		CRC YAML¹¹
APA Integration from D1		APA - /POST (Product Bundle Agreement API)¹²
CRC Integration from D1		POST /createCurrentAccount (Active IBAN)¹³
Product Order D1 OOB		Product Ordering Management Subdomain - Digital Experience Platform - Confluence AT - Production¹⁴
Aggregation Pattern¹⁵		Aggregation Pattern¹⁶

⁹ <https://confluence/display/DEP/Camunda+Step+Integration+Pattern>

¹⁰ https://amdocs.sharepoint.com/:u:/r/sites/ABN-AMRO2/Shared%20Documents/AMRO%20Project/Solution/External%20Systems/APA%20-%20ProductBundleAgreementAPI/ProductBundleAgreementAPI_v2.yaml?csf=1&web=1&e=dkdF1e

¹¹ <https://confluence/download/attachments/516556717/Current%20Account%20API.yaml?api=v2&modificationDate=1740659064732&version=1>

¹² <https://confluence/pages/viewpage.action?pageId=529564694>

¹³ <https://confluence/pages/viewpage.action?pageId=516556717>

¹⁴ <https://confluence/display/DEP/Product+Ordering+Management+Subdomain>

¹⁵ <https://confluence/display/DEP/Aggregation+Pattern>

¹⁶ <https://confluence/display/DEP/Aggregation+Pattern>

Document ID	Type	Document Name
Dependency Pattern ¹⁷		Dependency Pattern ¹⁸
Order History Tracking Events ¹⁹		Order History Tracking Events ²⁰
Self Healing Subdomain ²¹		Self Healing Subdomain ²²

7.1 Product Catalog

Characteristics: PA

Property	Value
Default Currency	EURO
System	CRC
Account Type	CURRENT_ACCOUNT

¹⁷ <https://confluence/display/DEP/Dependency+Pattern>

¹⁸ <https://confluence/display/DEP/Dependency+Pattern>

¹⁹ <https://confluence/display/DEP/Order+History+Tracking+Events>

²⁰ <https://confluence/display/DEP/Order+History+Tracking+Events>

²¹ <https://confluence/display/DEP/Self+Healing+Subdomain>

²² <https://confluence/display/DEP/Self+Healing+Subdomain>

7.2 D1 New Operations

7.2.1 AABNL Product Ordering Management MS

7.2.1.1 POST /....

7.2.1.1.1 Description

The **Activate Resource** capability will be used to **activate resource by type** and associate it with the **corresponding product** in the order.

The **default resource type** is **IBAN**, and in the future, additional resource types may be introduced based on system requirements.

7.2.1.1.2 Technical Description

The capability will perform the following:

- Search within the product structure for the **IBAN resource**
 - Characteristic with **resource type = IBAN** (property definition)
- Retrieve the necessary integration property values (**System & Account Type**)
- Invoke the integration accordingly

For **CRC System**, **DigitalONE** will invoke the **CRC reservation API** via **CRC Central Banking Account MS** to reserve the IBAN, passing the following details:

- **Customer ID** associated with the order
- **Account Type** value as specified by the property in **CatalogONE** (e.g. Account Type=**CURRENT_ACCOUNT**)

Once the IBAN is successfully reserved, the IBAN value will be persisted on the **Payment Account product**.

If the resource required, already exists on the **Payment Account**, it will be returned without invoking CRC reservation operation.

If the product order is not in draft state, reservation is not allowed.

The APA instance ID will be stored in the external ID object.

CRC Central Banking Account MS - will include 2 operations: reserve (used by OC) and activate (used by OH)

support partial success, error message - list of order item id which success/fails

7.2.1.1.3 Main Input

Attribute	Mandatory/ Optional	Description	Additional Information
Product Order ID	Mandatory		
Resource Type	Optional		Default = "IBAN"

7.2.1.1.4 Main Output - maybe not needed

Attribute	Description	Additional Information
Resource List []		
Order Item	The Order item ID	
Resource Number	The Resource Number (e.g. IBAN)	e.g. NL81ABNA0456279342
Type	The Type of the Resouce	e.g. IBAN